SYLLOGISM

All, No, Some: Statements and Conclusions

OAVS TGT | Reasoning (Verbal) | Topic 8 of the syllabus list

Syllabus line (Reasoning, 30 Marks): “The Verbal Reasoning topics like Series, Ranks, Direction, Arrangement, Coding, Decoding, Analogy and Classification/Odd Pair, **Syllogism** and Statement Conclusion.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). All 9 syllogism questions found in these papers are solved here, and every answer was checked by an exhaustive model check (all Venn cases).

What is inside

Ch.	Title	What you get
1	PYQ Analysis	Counts, patterns, priority, index of all 9 PYQs
2	Foundation	Four statement types with Venn diagrams, conversions, combination table, rules for ‘follows’, ‘can be’, either–or
3	Solved PYQs by Type	All–All chains, statements with No, same-subject statements, ‘can be’ conclusions
4	Shortcuts, Traps and Scope	Speed methods; what belongs to Statement–Conclusion
5	Practice Set	22 original questions with answer key
6	Quick Revision	One-page cheat sheet

SCOPE: Statement–Conclusion (statements about situations, courses of action, eligibility rules) is a different type, listed next in the syllabus line. Chapter 4 lists exactly which PYQs belong to it.

CHAPTER 1

PYQ Analysis

Syllogism: about 2 questions per paper (none in English 2022)

Paper	Q. numbers	Count	What was asked
English 2019	35, 38	2	Some + All statements; two conclusions each
English 2022	none	0	(Only statement–conclusion questions)
Mathematics 2024	81, 87, 88	3	Three statements each; 'All', 'No', 'Some' mixed
Science 2024	78, 80, 100	3	Three statements each; two 'can be' conclusions
Social Studies 2023	24	1	Three statements; three conclusions
Total		**9**	**7 of 9 are from the 2023 and 2024 papers; each has 2 or 3 conclusions**

Pattern	PYQs	Priority
'All' chains and 'All' + 'No' (conclusions follow by transitivity)	5	★★★ Must
'Some' + 'All' (Some A are B, All B are C → Some A are C)	2	★★★ Must
Same-subject statements (All O are X, All O are Y)	1	★★ High
'Can be' conclusions (possibility)	2	★★★ Must
Expected, not yet asked: Some … are not, either–or pairs, four-statement sets	0	★★ Expected

PYQ INSIGHT: The 'converse' trap is the most repeated distractor. Maths 2024 Q81 (II: 'Some trousers are watches'), Q87 (I: 'No coat is a jumper') and Science 2024 Q78 (I: 'No horses are ants') all give a conclusion that looks natural but is only **possible**, not certain.

PYQ INSIGHT: 'Can be' conclusions appeared twice in 2024 (Science Q80, Q100). They follow unless the statements **forbid** them. This is a different rule from 'are' conclusions, covered in 2.5.

Index of the 9 PYQs

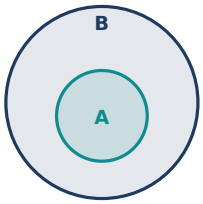
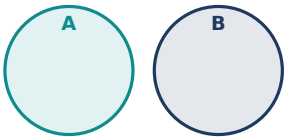
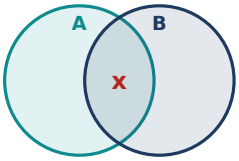
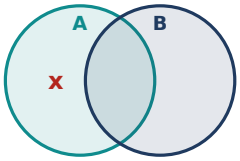
Paper / Q	Statements (short)	Conclusions	Ans	See
Eng 2019 Q35	Some bats are sticks; All sticks are rockets	1: Some bats are rockets; 2: No stick is bat	Only 1	3.1
Eng 2019 Q38	Some autos are scooters; All scooters are cycles	1: Some autos are cycles; 2: No scooter is auto	Only 1	3.1
Maths 2024 Q81	All trousers are pants; All pants are shirts; Some pants are watches	I: Some shirts are watches; II: Some trousers are watches	Only I	3.1
Maths 2024 Q87	All shirts are jackets; All jackets are coats; No jacket is a jumper	I: No coat is a jumper; II: No shirt is a jumper	Only II	3.2
Science 2024 Q78	All ants are socks; No parrots are socks; All horses are socks	I: No horses are ants; II: No ants are parrots	Only II	3.2
SST 2023 Q24	All utensils are spoons; All plates are buckets; No spoon is a bucket	I, II, III	All follow	3.2
Maths 2024 Q88	All ovens are speakers, boxes and curtains	I: All speakers are curtains; II: Some curtains are boxes	Only II	3.3
Science 2024 Q80	Some questions are answers; Some answers are options; All options are paragraphs	I: Some paragraphs can be questions; II: Some options can be questions	Both	3.4
Science 2024 Q100	Some harmoniums are keyboards; No keyboard is a piano; All pianos are instruments	I: Some keyboards can be instruments; II: Some instruments are harmoniums	Only I	3.4

CHAPTER 2

Foundation

Everything you need before solving

2.1 The four statement types

			
All A are B A is inside B	**No A is B** A and B do not touch	**Some A are B** A and B overlap (x = a member in both)	**Some A are not B** x = a member of A outside B

2.2 Conversions (what is always true)

Statement	Converse (swap)	Valid?
All A are B	Some B are A	**Yes** (limited converse). 'All B are A' is **not** valid.
No A is B	No B is A	**Yes**
Some A are B	Some B are A	**Yes**
Some A are not B	—	**No converse**

NOTE: Assumption used throughout: every group named in a statement has at least one member, and statements are true even if they contradict real facts ('All pants are shirts'). This is what makes 'All plates are buckets' give 'Some buckets are plates' (SST 2023 Q24, III).

2.3 Two-statement combination table

First	Second	Conclusion	PYQ
All A are B	All B are C	All A are C; Some C are A	Maths 2024 Q81
Some A are B	All B are C	Some A are C; Some C are A	Eng 2019 Q35, Q38
All A are B	No B is C	No A is C; No C is A; Some C are not A	Maths 2024 Q87; SST 2023 Q24
No A is B	All C are B (or All B are C)	No A is C (when C is inside B); Some C are not A (when C contains B)	Science 2024 Q78
Some A are B	No B is C	Some A are not C	—
All A are B	Some B are C	**No conclusion** (C may sit outside A)	Maths 2024 Q81 (II)
Some A are B	Some B are C	**No conclusion**	Science 2024 Q80
All A are B	All C are B	**No conclusion** (middle term not shared)	Science 2024 Q78 (I)
All O are X	All O are Y	Some X are Y; Some Y are X	Maths 2024 Q88

2.4 Method (Venn check)

Step	Do this
1	Draw the statements **as loosely as possible**: do not nest or merge circles unless a statement forces it.
2	Test each conclusion on the loosest drawing. A conclusion **follows** only if it is true in **every** way the circles can be drawn.
3	A 'can be' conclusion follows if there is **at least one** legal drawing where it is true (and no statement forbids it).

Step	Do this
4	If two conclusions are a complementary pair and neither follows alone, apply the either–or rule (2.6).

2.5 ‘Can be’ conclusions

Conclusion wording	Follows when…	Example
Some A can be B	no statement says No A is B or ties A completely away from B	Some paragraphs can be questions (Science 2024 Q80)
Some A are B (definite)	the statements force the overlap in every drawing	Some shirts are watches (Maths 2024 Q81)

2.6 Either–or rule

When two conclusions are contradictory (‘Some A are B’ with ‘No A is B’, or ‘All A are B’ with ‘Some A are not B’) and **neither follows alone**, but one of them must be true, the answer is **‘Either I or II follows’**. If one of them follows definitely (Eng 2019 Q35), the either–or rule is not used.

CHAPTER 3

Solved PYQs by Type

Venn reasoning, step by step

3.1 ‘Some’ and ‘All’ chains

PYQ Eng 2019 Q35 Some + All

Q. Statements: Some bats are sticks. All the sticks are rockets. Conclusions: 1) Some bats are rockets 2) No stick is bat. A. Either 1 or 2 B. Only 1 C. Neither D. Only 2

Step 1. Bats and sticks overlap, and the sticks sit inside rockets, so the bats that are sticks are rockets: Some bats are rockets.
Step 2. ‘No stick is bat’ contradicts the first statement. Since conclusion 1 follows, the either–or rule does not apply.
Answer: Only conclusion 1 follows (Option B)

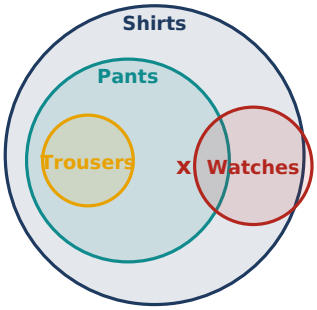
PYQ Eng 2019 Q38 Same pattern

Q. Statements: Some autos are scooters. All scooters are cycles. Conclusions: 1. Some autos are cycles 2. No scooter is auto.

Step 1. Some autos are scooters, and all scooters are cycles, so those autos are cycles: Conclusion 1 follows.
Step 2. ‘No scooter is auto’ contradicts ‘Some autos are scooters’.
Answer: Only conclusion (1) follows (Option D)

Maths 2024 Q81

All trousers are pants. All pants are shirts. Some pants are watches. Conclusions: (I) Some shirts are watches. (II) Some trousers are watches.



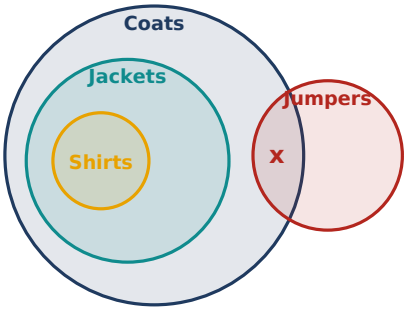
Conclusion	Test	Verdict
I. Some shirts are watches	Watches overlap pants (the x); pants are inside shirts, so that x is also in shirts.	**Follows**
II. Some trousers are watches	The overlap can be in pants outside trousers. Watches need not touch trousers.	Does not follow

RESULT: Answer: Only I (Option A). The x in the diagram sits where watches meet pants but not trousers, which shows II fails.

3.2 Statements with ‘No’

Maths 2024 Q87

All shirts are jackets. All jackets are coats. No jacket is a jumper. Conclusions: (I) No coat is a jumper. (II) No shirt is a jumper.



Conclusion	Test	Verdict
I. No coat is a jumper	Jumpers can overlap coats outside jackets (the x). So some coats can be jumpers.	Does not follow
II. No shirt is a jumper	Shirts are inside jackets, and jackets do not touch jumpers.	**Follows**

RESULT: Answer: Only II (Option C). 'No jacket is a jumper' does not mean 'no coat is a jumper': the coat circle is larger than the jacket circle.

Science 2024 Q78

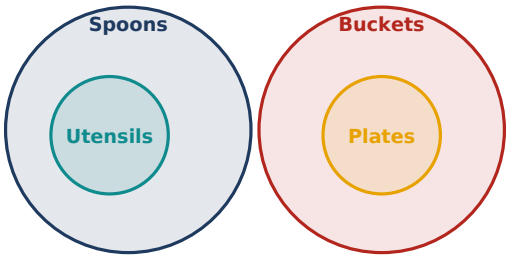
All ants are socks. No parrots are socks. All horses are socks. Conclusions: (I) No horses are ants. (II) No ants are parrots.

Conclusion	Test	Verdict
I. No horses are ants	Horses and ants are both inside socks. They may overlap completely or partly.	Does not follow
II. No ants are parrots	Ants are inside socks; parrots are outside socks entirely.	**Follows**

RESULT: Answer: Only II (Option 4). Two groups inside the same set (horses and ants inside socks) tell you **nothing** about each other.

SST 2023 Q24

All utensils are spoons. All plates are buckets. No spoon is a bucket. Conclusions: I. No utensil is a bucket. II. No plate is a utensil. III. Some buckets are plates.



Conclusion	Test	Verdict
I. No utensil is a bucket	Utensils are inside spoons; spoons do not touch buckets.	**Follows**
II. No plate is a utensil	Plates are inside buckets, utensils inside spoons; the two big circles are separate.	**Follows**
III. Some buckets are plates	All plates are buckets, so at least the plates are buckets (limited converse).	**Follows**

RESULT: Answer: all three follow (Option 4). Three separate groupings; every conclusion stays inside one pair of circles.

3.3 Same-subject statements

Maths 2024 Q88

All ovens are speakers. All ovens are boxes. All ovens are curtains. Conclusions: (I) All speakers are curtains. (II) Some curtains are boxes.

Conclusion	Test	Verdict
I. All speakers are curtains	Ovens sit inside all three groups, but speakers can extend outside curtains.	Does not follow
II. Some curtains are boxes	Ovens are inside curtains and inside boxes, so at least the ovens are in both.	**Follows**

RESULT: Answer: Only II (Option A). Several 'All X are …' statements with the same subject give 'Some … are …' between the predicates.

3.4 'Can be' conclusions

Science 2024 Q80

Some questions are answers. Some answers are options. All options are paragraphs. Conclusions: (I) Some paragraphs can be questions. (II) Some options can be questions.

Conclusion	Test	Verdict
I. Some paragraphs can be questions	Questions overlap answers; answers overlap options; options are inside paragraphs. A question that is an answer that is an option is a paragraph: possible, and nothing forbids it.	**Follows**

Conclusion	Test	Verdict
II. Some options can be questions	The same chain makes some options be questions possible; no statement says No question is an option.	**Follows**

RESULT: Answer: Both follow (Option 2). ‘Can be’ asks only whether it is **possible** under the statements.

Science 2024 Q100

Some harmoniums are keyboards. No keyboard is a piano. All pianos are instruments. Conclusions: (I) Some keyboards can be instruments. (II) Some instruments are harmoniums.

Conclusion	Test	Verdict
I. Some keyboards can be instruments	Keyboards avoid pianos but can be other instruments; nothing says No keyboard is an instrument.	**Follows**
II. Some instruments are harmoniums	Harmoniums touch keyboards, and instruments contain pianos. Keyboards need not be instruments, so harmoniums need not be either.	Does not follow

RESULT: Answer: Only I (Option 1). Notice the difference between ‘are’ (certain) in II and ‘can be’ (possible) in I.

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and what belongs to Statement–Conclusion

- **Combination table (2.3):** check which row your two statements match; no row = no conclusion.
- **No middle term, no conclusion:** both statements must share the same middle word.
- **'All' inside the same set tells nothing** about the two small groups (Science 2024 Q78, I).
- **Converse of 'All':** 'All A are B' gives 'Some B are A', not 'All B are A'.
- **'No' goes only as far as the circle:** 'No jacket is a jumper' does not give 'No coat is a jumper'.
- **'Can be':** follows unless a statement says No or ties the groups apart.
- **Contradicting a statement** means the conclusion cannot follow (Eng 2019 Q35 and Q38, conclusion 2).

Trap	How to avoid it
Using real-world facts	Statements are true even if absurd ('All pants are shirts'). Never use outside knowledge.
Treating 'Some' as 'all' or 'most'	'Some' means at least one and possibly all.
Concluding from two 'Some' statements	Some + Some gives no conclusion.
Reading 'can be' as 'are'	'Can be' is weaker: possibility is enough.
Using either–or when one follows	Either–or applies only when **neither** follows alone and the pair is complementary.
Reading a conclusion on a different pair of words	Both words must appear in the statements for a conclusion to follow.

Scope: PYQs that belong to Statement–Conclusion

PYQ	Why it is not here
Eng 2022 Q33; SST 2023 Q25	Statement with conclusions about opinions or causes: Statement–Conclusion
Eng 2022 Q50	Rules / criteria applied to candidates: Statement–Conclusion
Maths 2024 Q97	Statement with courses of action: Statement–Conclusion
Eng 2019 Q45	Eligibility criteria and referral rules (a long 'rule application' question): Statement–Conclusion

CHAPTER 5

Practice Set

22 original questions; answers on the next page

Decide which conclusions follow. Take the statements as true. Assume every named group has members.

Q	Statements	Conclusions
1	All pens are books. All books are bags.	I. All pens are bags. II. Some bags are pens.
2	Some cats are dogs. All dogs are birds.	I. Some cats are birds. II. All birds are dogs.
3	No tree is a flower. Some flowers are leaves.	I. Some leaves are not trees. II. No leaf is a tree.
4	All chairs are tables. No table is a bed.	I. No chair is a bed. II. Some beds are not chairs.
5	Some books are pens. Some pens are pencils.	I. Some books are pencils. II. Some pencils are books.
6	All A are B. All C are B.	I. All A are C. II. Some C are A.
7	All roses are flowers. Some flowers are red.	I. Some roses are red. II. No rose is red.
8	Some pens are cups. All cups are plates.	I. Some plates can be pens. II. No pen is a plate.
9	All lions are tigers. All tigers are wolves. No wolf is a deer.	I. No lion is a deer. II. Some wolves are lions. III. Some deer are tigers.
10	Some nails are hammers. All hammers are saws. Some saws are drills.	I. Some nails are saws. II. Some drills are hammers. III. Some saws are nails.
11	All rivers are lakes. All rivers are seas.	I. Some lakes are seas. II. All lakes are seas.
12	All doors are windows. Some windows are walls. No wall is a roof.	I. Some windows are not roofs. II. No door is a roof. III. Some doors are walls.
13	Some tables are chairs. No chair is a sofa.	I. Some tables can be sofas. II. Some sofas are not chairs.
14	All pens are pencils. No pencil is a rubber.	I. Some pens can be rubbers. II. No rubber is a pen.
15	All cats are animals. Some animals are not dogs.	I. Some cats are not dogs. II. Some dogs are not animals.
16	All balls are bats.	I. Some bats are balls. II. All bats are balls.
17	No fruit is a metal.	I. No metal is a fruit. II. Some fruits are not metals.
18	All A are B. All B are C. All C are D.	I. All A are D. II. Some D are A. III. All D are A.
19	Some A are B. All B are C. No C is D.	I. Some A are not D. II. Some C are A. III. Some B are not D.
20	No poet is a singer. All singers are dancers.	I. Some dancers are not poets. II. No dancer is a poet.
21	Some fruits are red. Some red things are sweet.	I. Some fruits are sweet. II. Some sweet things can be fruits.
22	Some men are doctors.	I. All men are doctors. II. Some men are not doctors.

Answer key with reason

Q	Answer	Reason
1	Both	All A are B, All B are C → All A are C; Some C are A (converse)
2	Only I	Some + All → Some cats are birds; All birds are dogs is the reverse and is not valid
3	Only I	No A is B, Some B are C → Some C are not A; ‘No leaf is a tree’ is too strong
4	Both	All chairs are tables, no table is a bed → No chair is a bed; beds exist and none is a chair → some beds are not chairs
5	Neither	Some + Some gives nothing (‘Some books can be pencils’ would be only possible)
6	Neither	Both circles inside B, no relation between A and C
7	Either I or II	Contradictory pair; neither follows alone, one must be true
8	Only I	Some pens are cups, cups are inside plates → some pens are plates; II contradicts this
9	I and II only	Lions inside tigers inside wolves; deer are outside wolves; III contradicts
10	I and III only	Some nails are hammers, hammers inside saws → Some nails are saws; Some saws are nails; II is not certain
11	Only I	Rivers inside both lakes and seas → Some lakes are seas; All lakes are seas is not certain
12	Only I	Windows overlap walls, walls avoid roofs → Some windows are not roofs; II and III are not certain
13	Both	Tables can include sofas outside chairs (possible); sofas are not chairs, so some sofas are not chairs
14	Only II	Pens are inside pencils which avoid rubbers, so ‘Some pens can be rubbers’ is forbidden; No rubber is a pen follows
15	Neither	Cats may or may not be dogs, and dogs need not lie outside animals; neither is certain
16	Only I	Limited converse of All; the full converse is not valid
17	Both	No A is B is symmetric; fruits exist, none is a metal → some fruits are not metals
18	I and II only	Transitive chain; Some D are A (converse); III is the full converse
19	All three	Some A are B and B inside C, C avoids D → A∩B avoids D; those A are C
20	Only I	Singers are dancers and no poet is a singer → Some dancers (the singers) are not poets; II is too strong
21	Only II	Some + Some gives nothing certain; ‘can be’ is possible
22	Either I or II	Contradictory pair (All X are Y vs Some X are not Y); neither follows alone

CHAPTER 6

Quick Revision

One page for the last day

Statements	Conclusion	PYQ
All A are B + All B are C	All A are C; Some C are A	Maths 2024 Q81
Some A are B + All B are C	Some A are C	Eng 2019 Q35, Q38
All A are B + No B is C	No A is C	Maths 2024 Q87 (II); SST 2023 Q24
Two groups inside the same set	Nothing about each other	Science 2024 Q78 (I)
All O are X and All O are Y	Some X are Y	Maths 2024 Q88 (II)
Some + Some; All + Some	No conclusion	Maths 2024 Q81 (II); Science 2024 Q80
'Can be' conclusion	Follows unless forbidden	Science 2024 Q80, Q100 (I)
Complementary pair, neither alone	Either I or II	Rule 2.6

Key rules

- Draw loosely; a conclusion follows only if true in every drawing.
- 'All A are B' gives 'Some B are A' (never 'All B are A').
- 'No A is B' gives 'No B is A'.
- A conclusion about a pair of words not linked by the statements does not follow.
- Never use real-world facts.

ROUTINE: Exam-day routine: draw the statements as loosely as possible; test each conclusion on the loosest drawing; check 'can be' and either–or last. 40 seconds per question.